

AUTOXR: REAL-TIME AR/VR VEHICLE CUSTOMIZATION

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Executive Summary:

1. Product and Customer Need – Pallavi

1.1 Preferences

AutoXR is an XR training and demo solution that helps explain vehicle features and complex tasks. We ran a Q15 choice-based conjoint with attributes including XR realism, where you can use it, collaboration, saved configurations, plans and monthly price.

Attribute importances (mean, Q15):

- Monthly price: 41.2 percent
- Collaboration features: 15.3 percent
- XR realism: 12.4 percent
- Where you can use it: 11.4 percent
- Plans: 10.2 percent
- Saved configurations: 9.5 percent

Price is the largest driver, but non-price features still represent almost 60% of the decision. Collaboration, realism and access together form a strong second layer of value, showing that customers care about **what the experience does** for them, not just how much it costs. This supports a **value-based, tiered pricing strategy** instead of competing as the cheapest option.

1.2 Problem statement and end benefit - (current content is from Prabhu)

Rescaled utilities quantify how current or weaker solutions feel to users.

XR realism (utilities):

- Level 3 (high detail, reflections): +37.42
- Level 2 (high detail, lighting): -36.88
- Level 1 (basic overlay): -0.54

Only full realism (Level 3) is clearly preferred. Mid-level realism and basic overlays sit at or below zero, indicating that “good enough” 3D is not very compelling when users have seen a higher standard.

Where you can use it (utilities):

- Showroom and home: +32.24
- Showroom only: -32.24

Showroom-only experiences carry a strong penalty. Users want the XR experience to extend beyond a single location so they can revisit it at home or in other contexts.

Together, these results suggest that current tools resemble **lower-realism, showroom-bound experiences**. They do help in the moment but do not provide ongoing, high-quality support. AutoXR’s opportunity is to deliver **high-realism XR that works both in the showroom and at home**, directly addressing this gap.

1.3 Target users and segmentation approach

Collaboration results are especially helpful for segmentation:

- Share link with dealer annotations: +20.30
- Share link with friends and family: +6.70
- No collaboration: -27.00

Users strongly dislike having no collaboration and value **dealer annotations** much more than social sharing. This points to **professional, guidance-oriented use cases** rather than purely social or entertainment-led ones.

Primary target segment:

- OEM dealerships and larger independent shops that:
 - Need to explain advanced features efficiently.
 - Want more consistent delivery and fewer post-sale questions.

In this segment, AutoXR acts as a **sales and education tool** that improves customer understanding and staff productivity. Social-only use cases and customers who prefer no XR solution are lower priority, given their lower utilities and likely lower willingness to pay.

2 Data and Methods

2.1 EVC market anchor (competitor + price + differentiators)

2.1.1 Closest competitor and benchmark price

2.1.2 Differentiators used to choose conjoint attributes

3 Primary Method: Choice-Based Conjoint (CBC) - Tanvir

Choice-Based Conjoint (CBC) is widely considered the "gold standard" of market research for product design and pricing. Unlike traditional surveys that ask respondents to rate features individually, CBC mimics the real-world psychology of making a purchase.

In a CBC study, respondents are shown a series of screens. Each screen displays a set of hypothetical product profiles, each with a specific combination of features (attributes) and a price. The respondents are asked for their choices.

The Output from respondents to choose between these bundles repeatedly, the analysis uses statistical modeling (typically Hierarchical Bayes) to derive:

- **Utility Scores:** The precise value (desirability) of every single feature level.
- **Importance Scores:** The attributes that drive decisions the most.
- **Market Simulation:** An analysis for the market to predict how the digital product's share changes if the feature or price is altered.

3.1 Why CBC for AutoXR

XR platforms are complex products where hardware limitations, software capabilities, and user comfort constantly compete. CBC is uniquely suited to solve the specific product management challenges in AutoXR for the following reasons:

3.2 Attributes and levels (3–4 features + price)

3.3 Sample and segmentation variables collected

3.4 Estimation approach and WTP conversion

4 Results & Discussion - Prabhu

4.1 Part-worth utilities and main drivers

Key mean rescaled utilities:

- **Plans:** Plan B (+17.14) is preferred; Plan C (-26.04) is disliked.
- **XR realism:** Level 3 (+37.42) is clearly best; Level 2 (-36.88) and Level 1 (-0.54) are much weaker.
- **Where you can use it:** Showroom and home (+32.24) vs showroom only (-32.24).
- **Saved configurations:** Up to 10 setups (+28.59) vs up to 3 (-28.59).
- **Collaboration:** Dealer annotations (+20.30), friends/family (+6.70), none (-27.00).
- **Price per month:** \$5 (+135.75), \$10 (-38.16), \$20 (-97.59).

A “best” configuration in this design combines Plan B, Level 3 realism, showroom and home access, 10 saved configurations, dealer annotations, and a lower price. A “worst” configuration stacks Plan C, lower realism, showroom-only access, few configs, no collaboration, and the highest price. The large gaps for realism and access show that **content quality and access flexibility are the main non-price drivers**, with collaboration and saved configurations adding further value.

4.2 Willingness-to-pay (WTP) for key features

Using the price attribute:

- Utility at \$5: 135.75
- Utility at \$20: -97.59

The difference is 233.34 utils for a 15-dollar change, so one util is worth about \$0.06 per month.

Approximate incremental willingness to pay:

1. **High realism XR (Level 3 vs Level 1):**
 - Utility difference: 37.96
 - WTP: about \$2.4 more per month
2. **Showroom and home vs showroom only:**
 - Utility difference: 64.48
 - WTP: about \$4.1 more per month
3. **Dealer annotations vs no collaboration:**
 - Utility difference: 47.30
 - WTP: about \$3.0 more per month

These values indicate that customers will pay **several dollars per month** for better realism, more usage occasions and guided collaboration. This supports a **basic vs premium tier** structure, where the premium plan bundles high-realism XR, showroom, and home access and dealer annotations at a higher price.

4.3 Simulations - Vinay

4.3.1 Take-up versus price for recommended bundle

4.3.2 Segment comparison (one key split)

5 Conclusion & Recommendations

5.1 Recommended configuration and price

5.2 Metrics (Goal / Driver / Guardrails)

5.3 Limitations

References

A Appendix A: Full EVC table and calculations

EVC for a Premium AutoXR Plan

Define a basic AutoXR plan as:

- XR realism: Level 1 (basic)
- Where you can use it: showroom only
- Collaboration: none
- Price: \$10 per month

Based on the WTP estimates:

- High realism XR (Level 3 vs Level 1): about \$2.4 per month
- Showroom and home access (vs showroom only): about \$4.1 per month
- Dealer annotations (vs none): about \$3.0 per month

Total differentiation value is roughly \$9.5 to \$9.6 per month.

Economic Value to the Customer for the fully loaded premium plan:

- $EVC = \text{baseline price} + \text{differentiation value}$
- $EVC = \$10 + \text{about } \9.6
- EVC is therefore about \$19.6 per month

This is the theoretical upper bound on what customers would pay for the premium configuration.

A practical launch strategy is to capture about **70-75%** of this value:

- 70% of \$19.6 is about \$13.7
- 75% of \$19.6 is about \$14.7

So a **recommended premium price** is in the **\$14 - \$15 per month** range, positioned clearly above a \$10 basic plan and leaving visible economic surplus for customers who benefit from higher realism, broader access, and expert annotations.